

POLICY BRIEF

Ending Gerrymandering: A One-Algorithm Solution

Track A, Paper 5

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Bottom line. A mathematical algorithm applied to public Census data—with no political input of any kind—produces congressional districts that are 22% more compact, closely proportional in partisan representation, and independently verifiable by any citizen in under ten minutes. One sentence of federal statute can make gerrymandering structurally impossible.

The Problem

Every ten years, after the Census, the party in power in most state legislatures redraws congressional district boundaries to ensure that party wins more seats than the voters it represents would otherwise produce—a practice called **gerrymandering**. In 2019, the U.S. Supreme Court held in *Rucho v. Common Cause* that federal courts are powerless to remedy partisan gerrymandering, leaving Congress as the only institution that can act [7]. Congress has not acted. The result: in state after state, mapmakers stretch districts into grotesque shapes, packing opposition voters into a few districts and cracking communities across many, to predetermine the outcome of the next decade of elections before a single vote is cast.

Rucho explicitly invited Congress to fix the problem. This brief proposes how.

The Solution

Replace the politician with an algorithm.

The algorithm works like splitting a pizza: make one cut through the middle of the state, dividing it into two equal-population halves. Cut each half in half. Repeat until the state has as many pieces as it has congressional seats. Every cut is guided by one rule—make it as short as possible—so districts follow natural geographic features instead of zigzagging to capture partisan voters.

The algorithm uses U.S. Census population counts and tract boundary geometries, both of which are public data [5]. It never sees voter registration files, election results, or incumbents' home addresses. **It cannot gerrymander because it lacks the information required to do so.** This is not a policy choice that a future legislature could reverse; it is a structural property of the algorithm.

The algorithm is a direct extension of *Huntington-Hill*—the mathematical method Congress already uses to apportion seats among states, mandated by statute since 1941 [6]. Extending the same principle from apportionment to boundary design requires one sentence of statute.

Two independent verifiers running the algorithm on the same Census data produce byte-identical district assignments, auditable through a SHA-256 hash chain. Any citizen can verify the result in under ten minutes.

What the Evidence Shows

Applied to all fifty states across the 2000, 2010, and 2020 Census cycles, the algorithm produces the following results:

- +22%** More compact than enacted 2020 maps, measured by the Polsby-Popper ratio of district area to perimeter squared, averaged across all 435 districts. Compact districts follow rivers and county lines rather than twisting through neighborhoods to sort voters. (*Paper B.2 [2]*)
- 7D / 7R** North Carolina—the state that has generated more federal gerrymandering litigation than any other in the past decade—receives a seven-Democratic, seven-Republican delegation under the algorithm, closely matching the state’s evenly divided electorate. No partisan data was used. (*Paper B.11 [1]*)
- 10 min** Any citizen with a laptop and an internet connection can download the open-source `redist` tool, fetch public Census data, run the Vermont verification fixture, and confirm the result themselves in under ten minutes. Full fifty-state replication takes approximately two hours. (*Replication Package A.4*)

Additional findings: the algorithm produces **more majority-minority districts** than enacted plans in states above the 42% minority-population threshold identified in Paper D.1 [4], and reduces the partisan efficiency gap by **62%** compared to enacted plans [3]—without any partisan fairness criterion in the algorithm specification.

Three Paths to Adoption

1. Federal statute.. The cleanest solution requires one sentence added to federal law:

Each State shall apportion its congressional districts using the Huntington-Hill recursive bisection algorithm applied to decennial Census redistricting data, subject to requirements of the Voting Rights Act.

This language is analogous to the statutory mandate for Huntington-Hill apportionment, which has operated without controversy since 1941. No independent commission is needed; no appointments, no budget, no institutional design problem. The algorithm is the commission.

2. State commission or legislative adoption.. States may adopt algorithmic redistricting by statute or ballot initiative without waiting for Congress. The `redist` tool already supports any state’s seat count and can be configured to respect existing statutory criteria (municipal boundaries, county lines, communities of interest). States that adopt algorithmic redistricting become natural laboratories, and their experience will inform the federal case.

3. Court-ordered remedy.. After a court finds a gerrymander under state constitutional standards (as courts have in Pennsylvania, North Carolina, and New York),

algorithmic plans provide a judicially administrable remedy. Rather than appointing a special master to draw a map— an ad hoc, unreviewable process—a court can order the parties to submit the Census data to `redist` and adopt the output. The SHA-256 audit chain makes the result tamper-evident and verifiable by any party or amicus.

Resources and Contact

Open-source code.. The full `redist` source code, including the METIS graph-partitioning library, is available at:

<https://github.com/giodl73-repo/REDIST>

The code is written in Rust and is freely available for inspection, audit, and independent replication under an open-source license.

Interactive dashboard.. District maps for all fifty states and all three census years (2000, 2010, 2020) are viewable in a browser-based dashboard bundled with the repository at `web/dashboard.html`. No software installation is required.

Research portfolio.. The full research portfolio—more than forty papers covering the algorithm, legal compliance, validation, and applications—is maintained at `research/` in the same repository. A non-technical visual guide is in Paper A.3. The reproducibility package is in Paper A.4.

Contact.. Questions, comments, and collaboration inquiries are welcome.

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References

- [1] Giovanni Della-Libera. Apportion regions: Bisection-based sub-state apportionment. *Working paper*, 2026. B.11 — NC 7D/7R near-proportional result.
- [2] Giovanni Della-Libera. Edge-weighted graph partitioning for geographic boundary optimization. *Working paper*, 2026. B.2 — +22% compactness improvement, 50 states.
- [3] Giovanni Della-Libera. Efficiency gap analysis of algorithmic redistricting plans. *Working paper*, 2026. C.5 — 62% partisan bias reduction, 50 states.
- [4] Giovanni Della-Libera. Voting rights act compliance through algorithmic redistricting. *Working paper*, 2026. D.1 — 42% VRA demographic threshold.

- [5] U.S. Census Bureau. 2020 census redistricting data (PL 94-171). <https://www.census.gov/programs-surveys/decennial-census/about/rdo/summary-files.html>, 2021.
- [6] U.S. Congress. Method of equal proportions (huntington-hill), codified at 2 u.s.c. sec. 2a, 1941. Statutory mathematical apportionment in use since 1941. Codified at 2 U.S.C. 2a.
- [7] U.S. Supreme Court. Rucho v. common cause, 588 u.s. 684 (2019), 2019. Partisan gerrymandering claims present political questions beyond the reach of federal courts.